

## Switch Template

1. Prepare Linux server
  - a. `sudo apt update`
  - b. `sudo apt upgrade -y`
  - c. `sudo apt install -y openvswitch-switch`
  - d. `sudo apt install -y net-tools`
  - e. `sudo apt install -y iperf3`
  - f. `sudo apt install -y lldpd`
2. Set hostname and timezone
  - a. `sudo hostnamectl set-hostname <switch name>`
  - b. `sudo nano /etc/hosts`
    - i. Replace loopback line with <switch name>
      1. `127.0.1.1 <switch name>`
  - c. `sudo timedatectl set-timezone Australia/Brisbane`
3. Enable Open vSwitch
  - a. `sudo systemctl enable --now openvswitch-switch`
4. Enable LLDP & CDP
  - a. `sudo nano /etc/default/lldpd`
    - i. Type:
      1. `DAEMON_ARGS="-c"`
    - ii. Save and exit (Ctrl + x)
  - b. `sudo mkdir -p /etc/lldpd.d`
  - c. `sudo nano /etc/lldpd.d/interface.conf`
    - i. Type:
      1. `configure system interface pattern ens12,ens13,ens14,vlan99`
  - d. `sudo systemctl enable --now lldpd`
  - e. `sudo systemctl restart lldpd`
5. Run:
  - a. `ip link set dev ens<port number> address 26:13:23:6a:<switch/AP last octet in IP address>:<port number>`
6. Configure default network ports and secure permissions on file
  - a. `sudo nano /etc/netplan/00-installer-config.yaml` (for macaddress – a=site & b=Brisbane)
    - i. Type:
      1. `network:`  
`version: 2`  
`renderer: networkd`  
  
`ethernets:`  
`ens3:`  
`dhcp4: no`  
`macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:01`  
`ens4:`  
`dhcp4: no`  
`macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:02`  
`ens5:`  
`dhcp4: no`  
`macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:03`  
`ens6:`  
`dhcp4: no`  
`macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:04`

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ens7:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:05
ens8:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:06
ens9:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:07
ens10:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:08
ens11:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:09
ens12:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:10
ens13:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:11
ens14:
  dhcp4: no
  macaddress: 26:13:23:6a:<switch/AP last octet in IP address>:12

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- b. `sudo chmod 600 /etc/netplan/00-installer-config.yaml`
  - c. `sudo netplan apply`
7. Disable waiting for networks to come online after restart
  - a. `sudo systemctl disable systemd-networkd-wait-online.service`
  - b. `sudo systemctl mask systemd-networkd-wait-online.service`
8. Create OVS Bridge and configure RSTP, Multicasting snooping and Fail Mode.
  - a. `sudo ovs-vsctl add-br br0`
  - b. `sudo ovs-vsctl set bridge br0 rstp_enable=true`
  - c. `sudo ovs-vsctl set bridge br0 mcast_snooping_enable=true`
  - d. `sudo ovs-vsctl set bridge br0 fail_mode=standalone`
9. Add Ports to Bridge so they are managed in OVS plus add description of device added to Port.  
This is for ports with end devices (ens3=port1 up to ens11=port9)
  - a. `sudo ovs-vsctl add-port br0 ens<port number> tag=<VLAN number> -- set interface ens<port number> external-ids:port-desc="<description of device>"`
    - i. repeat step above for each port
    - ii. place spare ports on vlan1 and add description of "spare port"
10. Add Ports to Bridge so they are managed in OVS plus add description of device added to Port.  
This is for ports in Trunk Mode (ens12=port10 to ens14=port12)
  - a. `sudo ovs-vsctl add-port br0 ens<port number> trunks=<VLAN number>,<VLAN number>,(repeat for all VLANs on that trunk) -- set interface ens<port number> external-ids:port-desc="connected to <switch/AP name>"`
11. Configure VLAN99
  - a. `sudo ovs-vsctl add-port br0 vlan99 tag=99 -- set interface vlan99 type=internal`
  - b. `sudo ip link set vlan99 up`
  - c. `sudo nano /etc/systemd/network/vlan99.network`
    - i. Type:
      1. [Match]  
Name=vlan99

[Network]

Address=<switch/AP IP address>/24

Gateway=192.168.99.1 (for BNE) or 192.168.199.1 (for site)

DNS=<same IP address as gateway>

DNS=8.8.8.8

d. `sudo systemctl enable systemd-networkd`

e. `sudo systemctl restart systemd-networkd`

12. Enable QoS to match and represent port speed (this example is for ens3 (port 1) which is normally set to 100Mbps. For ports between switches, set to 1Gbps)

a. `ovs-vsctl -- set port ens3 qos=@newqos \`

b. `-- --id=@newqos create qos type=linux-htb other-config:max-rate=100000000 \`

c. `queues=0=@q0 \`

d. `-- --id=@q0 create queue other-config:max-rate=100000000`

13. Create a static MAC address for VLAN99

a. `sudo ovs-vsctl set interface vlan99 mac="26:13:23:6a:99:<switch/AP last IP octet number>"` (use 'a' for site and 'b' for Brisbane)

14. Set the login banner

a. `sudo nano /etc/issue`

i. Type in banner

1. Type:

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b. `sudo nano /etc/motd`

i. Type in banner

1. Type:

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c. `Sudo chmod 644 /etc/issue /etc/motd`

15. Reboot

a. Reboot

16. Test connection

a. `ping -c 4 <ip address of other switches/AP/router>`